

Semester IV

Core-VIII: Computer Graphics

Course Objectives:

- To understand basic concepts of computer graphics.
- To learn techniques for creating basic graphical structures
- To learn different transformation techniques

Learning Outcomes:

Upon completion of this course, students will be able to:

- Know the use of different graphics systems
- Learn different algorithms to draw geometrical figures
- Learn various geometric transformation techniques
- Learn techniques for clipping

Unit I:

Computer Graphics: A Survey of Computer graphics, Overview of Graphics System: Video Display Devices, Raster-Scan Systems, Input Devices, Hard-Copy Devices, Graphics Software.

Unit II:

Graphics Output Primitives: Point and Lines, Algorithms for line, circle & ellipse generation, Filled-Area Primitives. Attributes of Graphics Primitives: Point, line, curve attributes, fill area attributes, Fill methods for areas with irregular boundaries.

Unit III:

Geometric Transformations (both 2-D & 3-D): Basic Geometric Transformations, Transformation Matrix, Types of transformation in 2-D and 3-D Graphics: Scaling, Reflection, shear transformation, rotation, translation. 2-D, 3-D transformation using homogeneous coordinates.

Unit IV:

Two-Dimensional Viewing: Introduction to viewing and clipping, viewing transformation in 2-D, viewing pipeline, Clipping Window, Clipping Algorithms: Point clipping, Line clipping and Polygon clipping.

Text Books:

- ✓ *Donald Hearn & M. Pauline Baker, "Computer Graphics with OpenGL", Pearson Education.*
- ✓ *Mathematical Elements for Computer Graphics, D. F. Rogers & J. A. Adams, MGH, 2/ed.*

Reference Books:

- ✓ *Computer Graphics principles & practice, Foley, Van Dam, Feiner, Hughes Pearson Education*
- ✓ *Computer Graphics by Zhigang Xiang, Roy A Plastic, McGraw-Hill*

Core-VIII Lab: Computer Graphics using OpenGL

1. Write a program to implement Bresenham's line drawing algorithm.
2. Write a program to implement mid-point circle drawing algorithm.
3. Write a program to clip a line using Cohen and Sutherland line clipping algorithm.
4. Write a program to clip a polygon using Sutherland Hodgeman algorithm.
5. Write a program to fill a polygon using Scan line fill algorithm.
6. Write a program to apply various 2D translation transformation.
7. Write a program to apply 2D object homogenous coordinates translation.
8. Write a program to apply various 2D rotation transformation.
9. Write a program to apply 2D object homogenous coordinates rotation.
10. Write a program to apply various 2D scaling transformation.
11. Write a program to apply 2D object homogenous coordinates scaling transformation.
12. Write a program to apply various 3D transformations on a 3D object and then apply parallel and perspective projection on it.